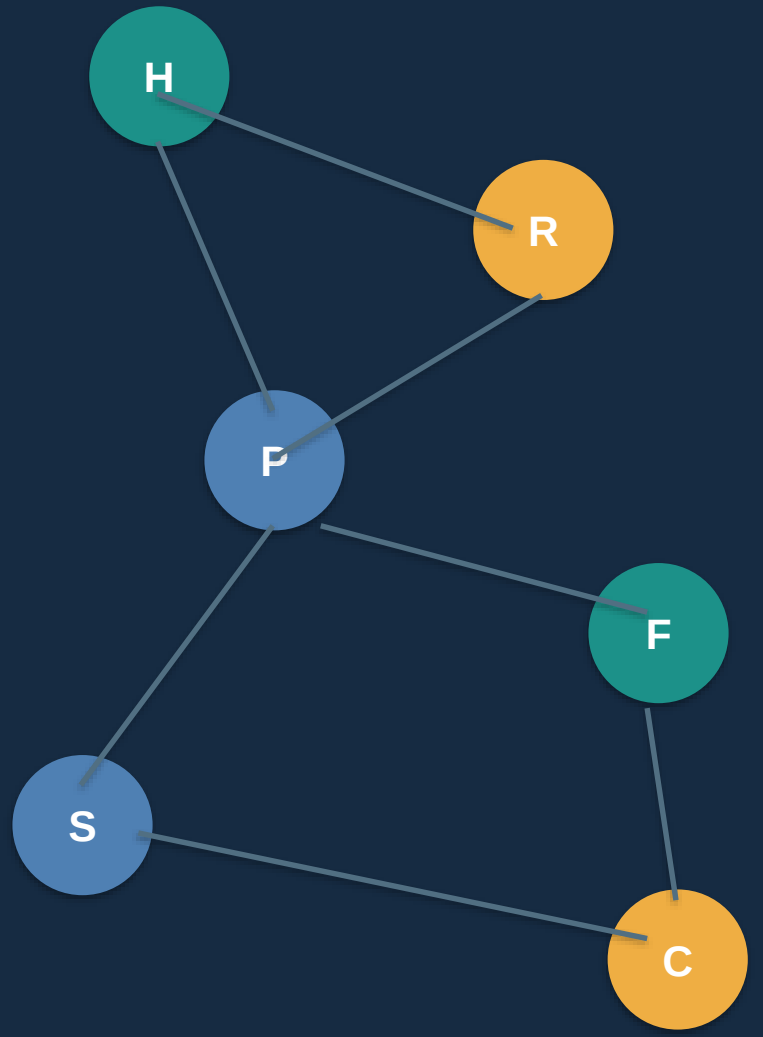
COMPUTER NETWORKS • COURSE PROJECT

SafeRoute

Adaptive Disaster-Resilient Communication Network Simulator

A network simulation that keeps critical messages moving when links or nodes fail.



Abhinav Sharma • Slot F1 • Team 11
Faculty: SREETHAR S

When a disaster breaks the network, what happens next?

The challenge is not only sending packets. It is keeping essential communication available.

An earthquake damages a communication link between an emergency centre and a hospital.

Ordinary traffic

Routine messages still compete for limited network capacity.

Critical traffic

An ambulance or hospital alert needs a fast, reliable route.

Network failure

A route may become unavailable, so traffic must find another path.

SafeRoute lets us trigger this situation, observe the network, and measure the response.

The problem SafeRoute explores

A controlled simulation helps us study routing, priority, failures, and recovery without real infrastructure.

Problem statement

In a disaster, links or nodes can fail while emergency messages still need to reach critical services. Static routes and congestion can delay delivery.

Project objective

Build an interactive network simulator that models topology, packet traffic, emergency priority, failures, rerouting, and recovery, then displays the resulting behaviour.

Scope: an educational software simulator, not a production emergency communication system.

From the dashboard to the simulation engine

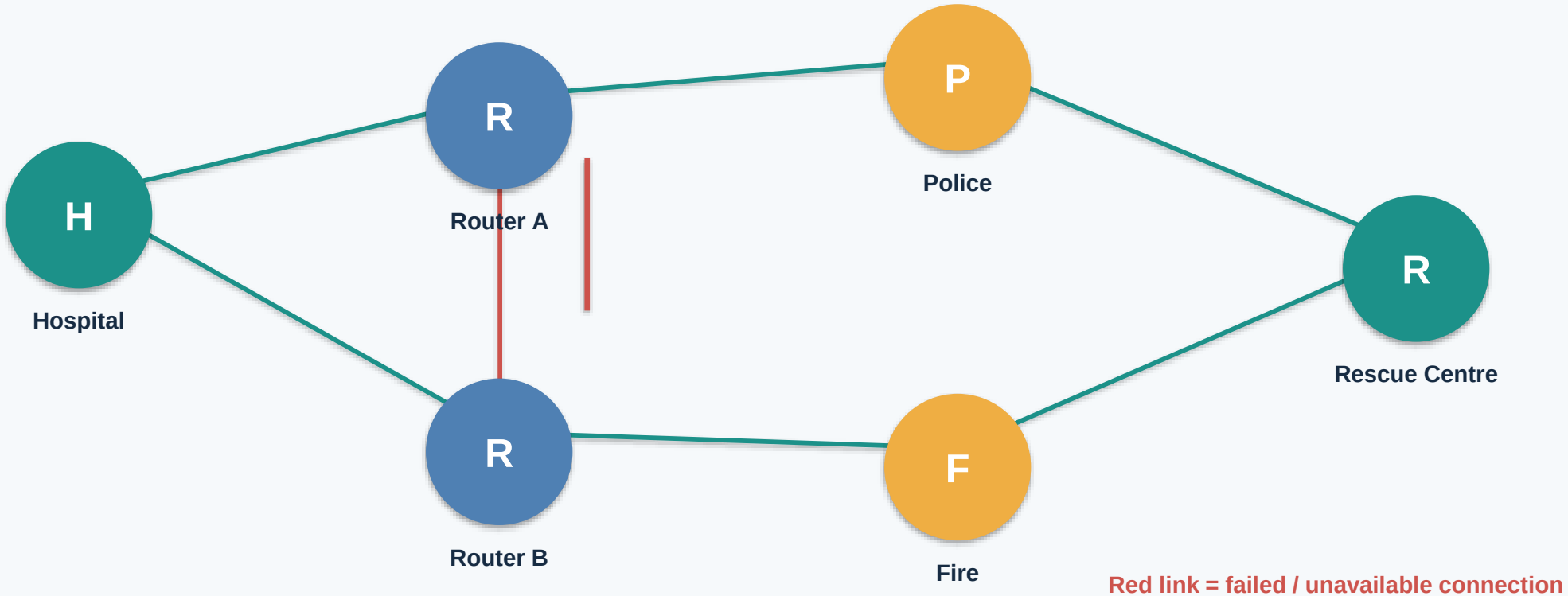
The interface sends commands to the API; the engine updates network state and returns metrics and events.



Key idea: the frontend visualizes the model; the engine is responsible for the network behaviour.

What does the simulated network look like?

Each node represents a service or network device. Each link represents a connection with capacity and state.



Example: if Router A–B fails, packets may use another available path through the network.

One experiment, from setup to observation

This is the sequence you can demonstrate live to your faculty.

1. **Build topology**
   Create nodes and links
2. **Generate traffic**
   Normal or emergency packets
3. **Run simulation**
   Engine advances in ticks
4. **Trigger a failure**
   Disable a node or link
5. **Reroute traffic**
   Find an available path
6. **Read outcomes**
   Metrics, events, delivery

Observe: Did the packet arrive? Was a route available? How did delay and congestion change?

Dijkstra: choose a low-cost available route

The simulator uses shortest-path routing over the links that are currently available.

Before failure

Hospital → Router A → Rescue Centre

The path has a lower total cost, so the engine can select it.

Failure occurs

A link becomes unavailable.

That edge must be excluded from route calculation; the old route may no longer be usable.

After failure

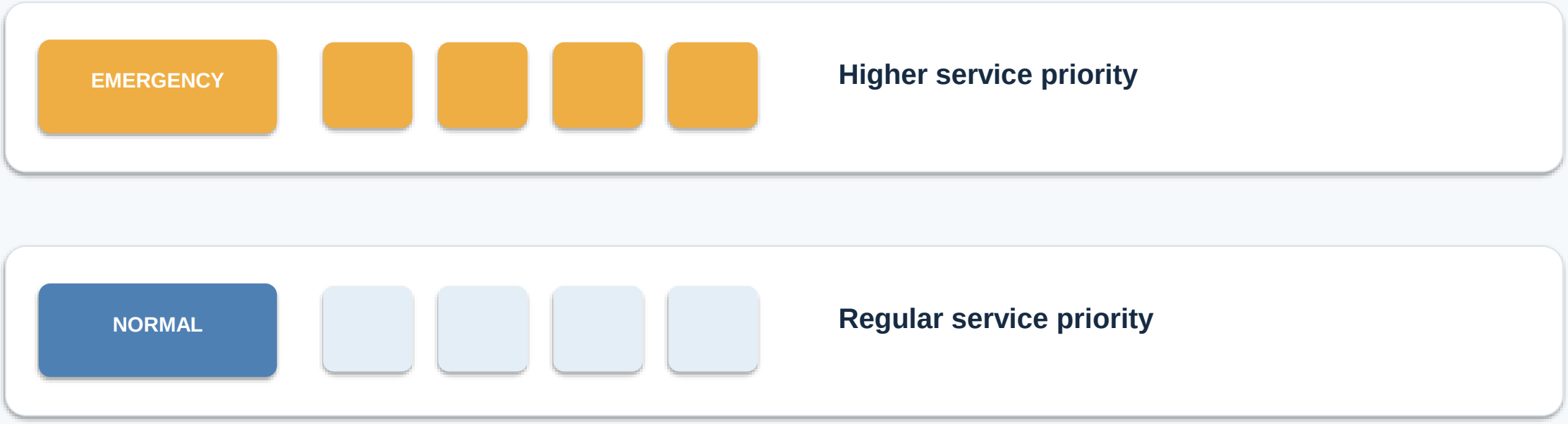
Hospital → Router B → Fire → Rescue Centre

If this alternate path is available, packets can continue along it.

Important: rerouting is possible only if an alternate connected path exists.

Not all packets have the same urgency

SafeRoute models emergency traffic separately so we can examine how priority affects network behaviour.



Discussion point: priority can help critical traffic, but it does not create bandwidth or guarantee delivery when every route is broken.

Disaster presets create repeatable scenarios

Use presets to trigger a set of failures and compare what happens to the network.

EARTHQUAKE

Multiple local nodes or links can become unavailable.

FLOOD

A region or group of links may be isolated.

CYCLONE

Several connections may degrade or fail.

CYBERATTACK

A scenario can model disrupted network components.

These are simplified, category-based presets for learning and comparison, not physical disaster predictions.

How do we judge whether the network is coping?

Metrics and event logs help explain the outcome instead of relying only on the animation.

Packets delivered

How many packets reached their destination?

Packets dropped

How much traffic could not be delivered?

Delay / latency

How long did delivery take in simulation ticks?

Link utilization

Which connections are carrying the most traffic?

Node / link state

Which components are active, failed, or recovered?

Event history

What happened, and in what sequence?

A useful comparison: baseline network vs. network after failure vs. network after recovery.

The technology behind SafeRoute

The project separates visual interaction, API handling, and network simulation logic.

Frontend

React 19 • Vite
React Flow for topology visualization

Backend

Node.js • Express
REST API and simulation controls

Engine + tests

JavaScript simulation classes
Node built-in test runner

Suggested verification commands

```
npm run dev
```

What the tests should cover

- Routing and path availability
- Packet delivery and failure handling
- Priority, reset, and recovery

What SafeRoute does, and what it does not claim

Being clear about the model makes the project easier to evaluate fairly.

It does

- Simulates nodes, links, and packets
- Demonstrates Dijkstra-based routing
- Models priority and component failures
- Shows metrics and event history

It does not

- Replace real emergency networks
- Emulate full TCP/IP behaviour
- Model every routing protocol detail
- Guarantee service if no route survives

The value is a transparent, repeatable environment for studying network resilience concepts.

TAKEAWAY

SafeRoute makes network resilience visible.

It lets us create a network, introduce traffic, break components, observe rerouting, and evaluate the outcome.

Thank you !!